conservation and restoration policies. Traditional data collection methods are labor-intensive and expensive, but novel artificial intelligence (AI) tools are transforming the field. For example, camera traps and multifrequency acoustic recording devices can remotely identify, photograph, and log animals. Underwater devices can produce 3-D models of coral reefs or audio recordings of fish sounds—analyzed by AI to determine reef health. AI can track animals and map habitat changes, and propose ecosystem restoration strategies.

Citizen science

A global army of “citizen scientists” can now input data onto smartphones via apps such as iRecord and eBird, which use AI to identify organisms using photos and sound recordings. This information is fed instantly to global databases, where AI analysis determines trends. Between 2014–24, over 1 million amateur naturalists added 1.9 billion bird records to Cornell’s eBird database. Cornell added the Merlin Sound ID app to the photo-identification facility in 2021, allowing users to identify and input data based on birds’ songs and calls. In 2024, more than 12 million people used this app. Ever more creative applications of AI will be an increasingly important tool to combat biodiversity loss.